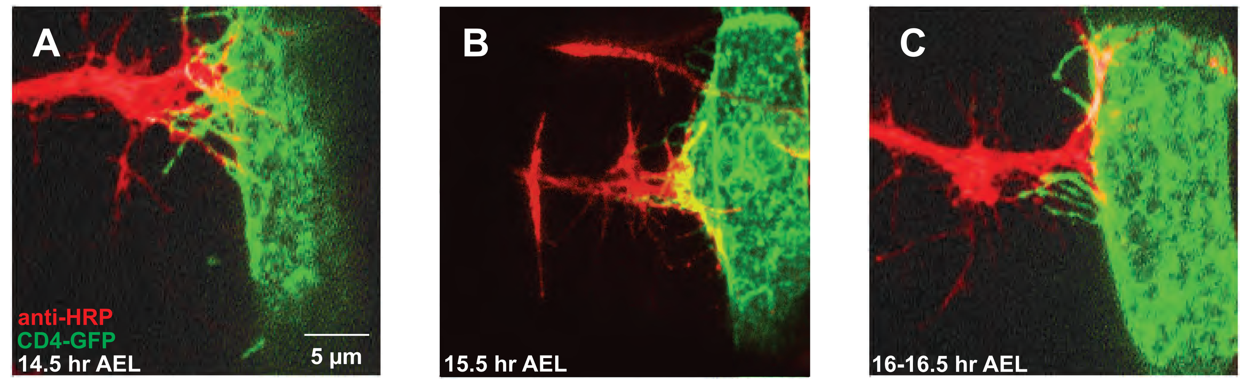
# Characterization of membrane markers for localization-based super-resolution imaging of filopodia at the neuromuscular junction in *Drosophila*

Melissa Inal*, Abhijit Marar *, Ruizhe Lin, Peter Kner and Daichi Kamiyama
Department of Cellular Biology, University of Georgia, Athens, GA, USA (*) contributed equally to this work

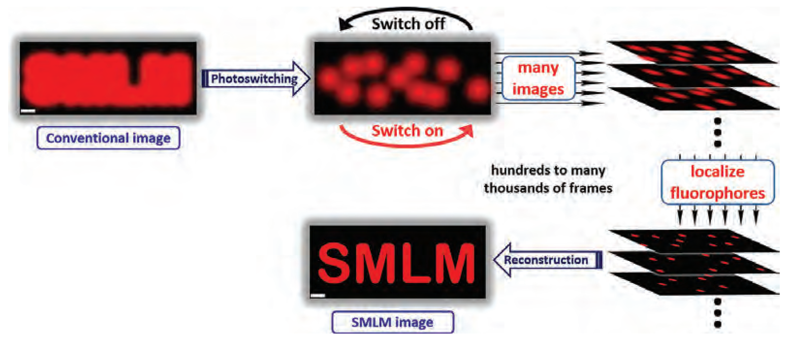
## Synaptogenesis during embryonic development

### *Investigating filopodia contacts at early synaptogenesis to uncover molecular mechanisms of recognition and recruitment of synaptic components*



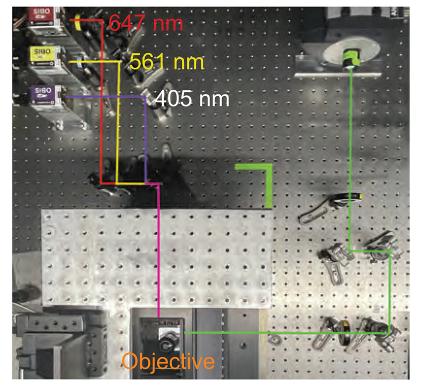

During synaptogenesis, the filopodia from the growth cone recognize the filopodia on the muscle forming fiopodia clusters (**A**), lamellipodia-like structure on muscle (**B**) and bifurcation on the growth cone (**C**).
AEL: after egg laying

### *Single Molecule Localization Microscopy (SMLM)*




Li H, Vaughan JC. Switchable Fluorophores for Single-Molecule Localization Microscopy. Chem Rev. 2018;118(18):9412-9454. doi:10.1021/acs.chemrev.7b00767

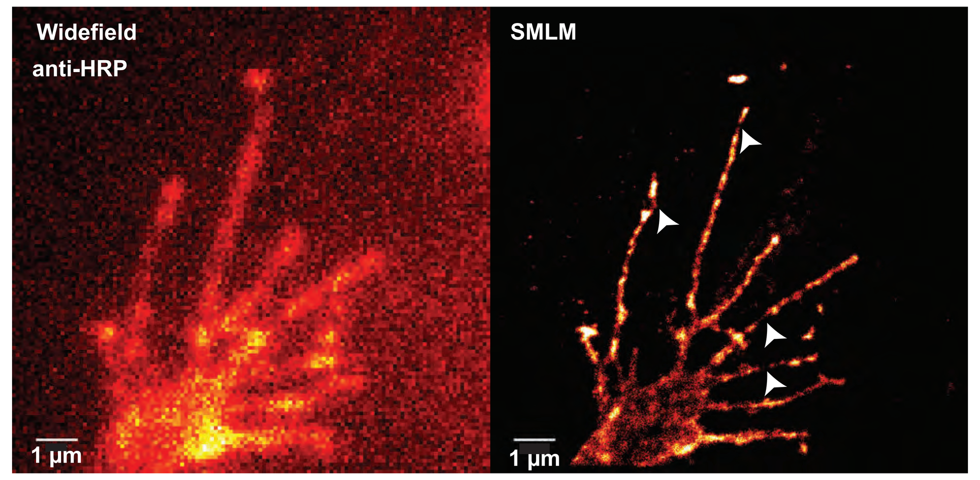
The schematic shows the flow of data acquisition and reconstruction for single molecule localization microscopy.




We use a custom built inverted microscope equipped with 647 nm, 561 nm (for switching off fluorophores), and 405 nm (for activation) laser lines.

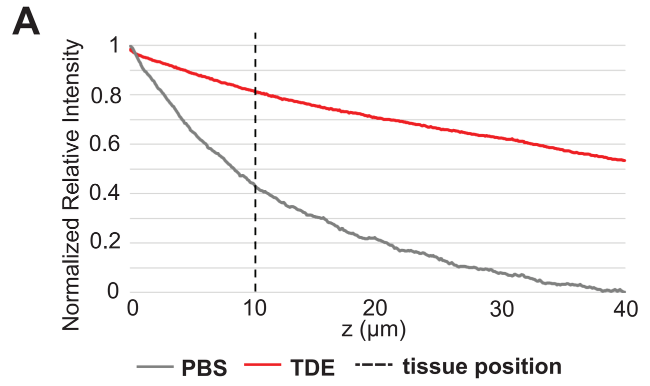
## Setting up for optimal imaging on the growth cone

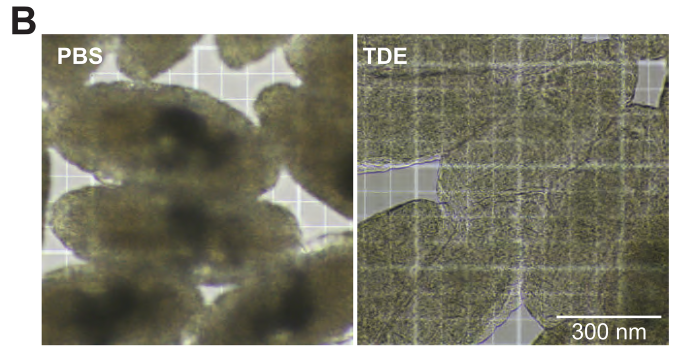
### *Preliminary growth cone image*




A farred dye shows promise for SMLM imaging of the aCC motor neuron at 15 hr AEL. Arrowheads point to gaps in labeling.

### *Improving signal sensitivity*

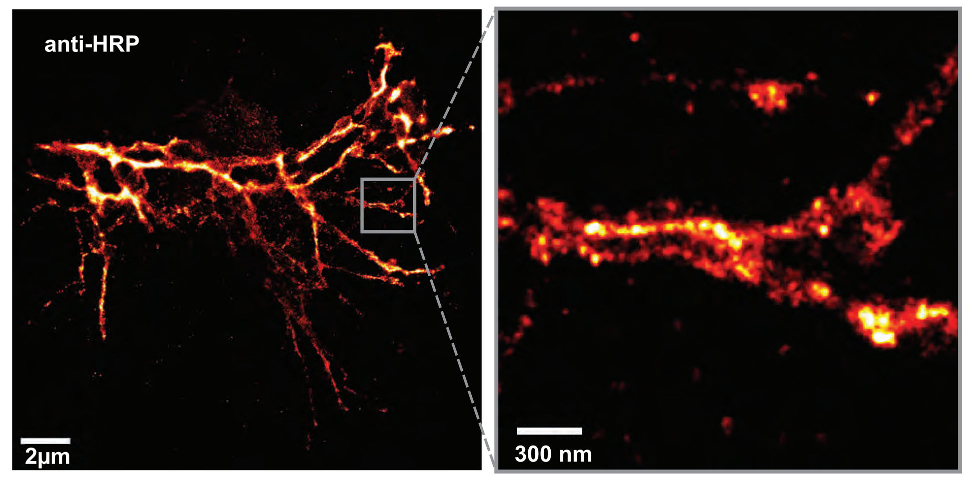



(**A**) Relative fluorescence intensity farred dye over change in depth (z) starting from coverslip (z=0) (**B**) TDE-based buffer renders embryos transparent.

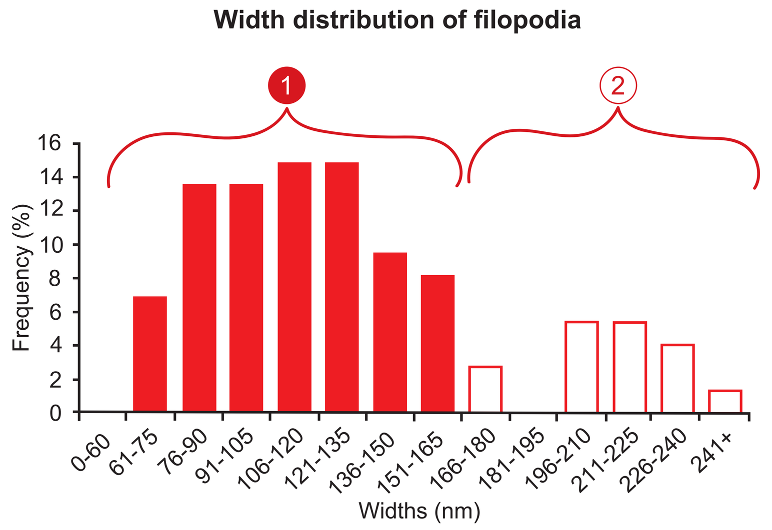
## Optimization allows for quantitative measurements

### *Optimized growth cone image*

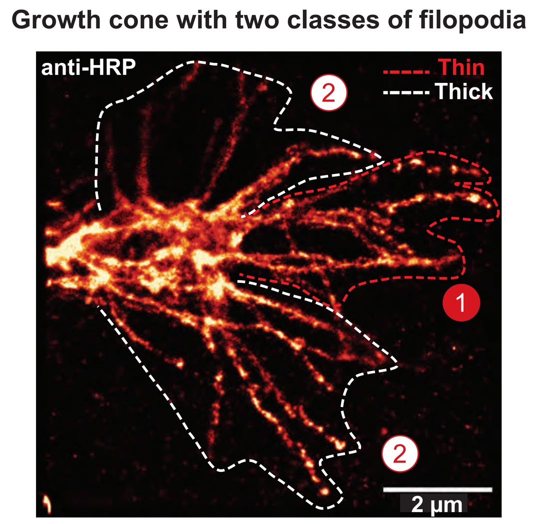



Optimized conditions on the ISNb growth cone at 15 hr AEL (left panel) allow us to resolve individual filopodia (right panel) with improved continuity in membrane labeling.

### *Classifying filopodia by width*



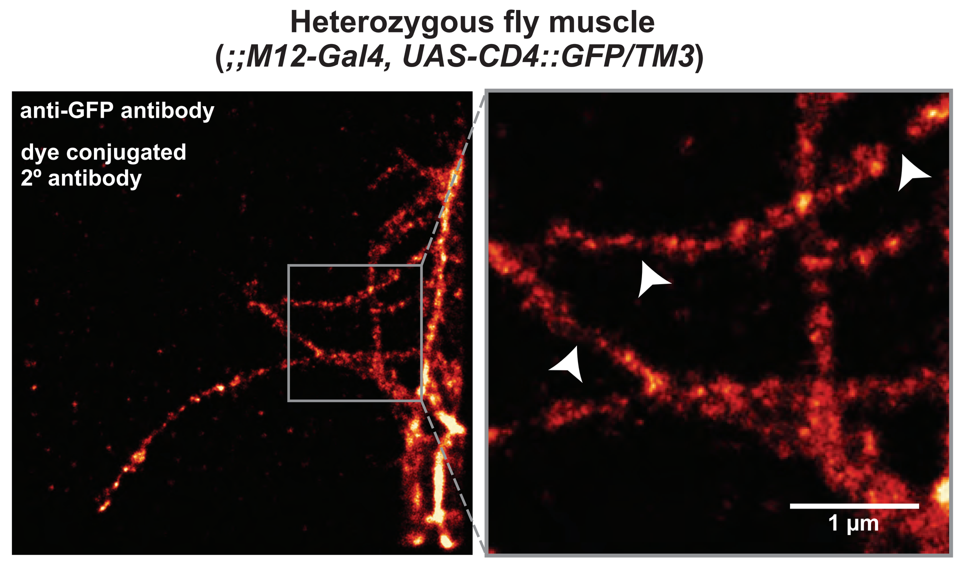

The width of individual filopodium was measured and plotted as a frequency histogram. Filopodia on the growth cone may have a bimodal distribution pattern for their widths.



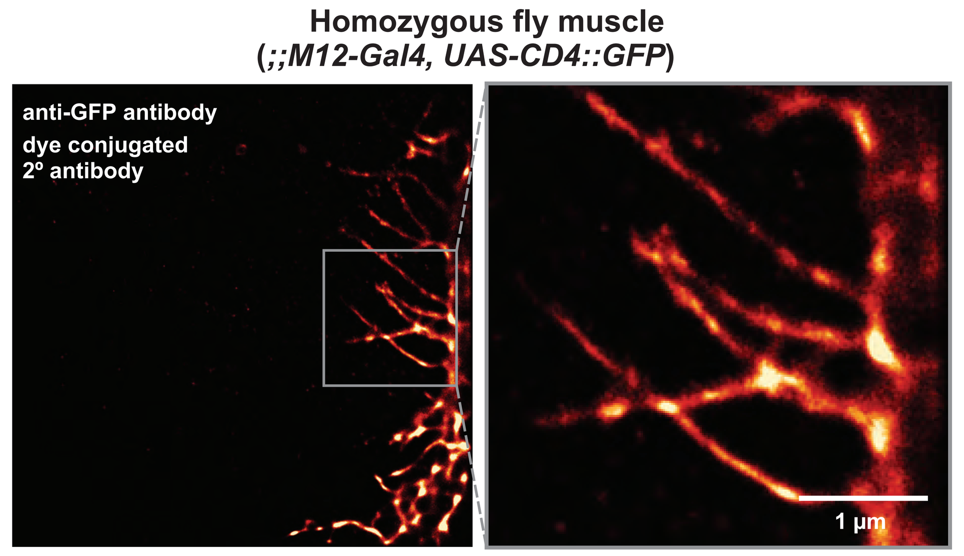

A representative image of ISNb growth cone shows two classes of filopodia (red and white dashed lines).

## Setting up for optimal imaging on the muscle

### *Improving muscle labeling increasing genetic dosage*

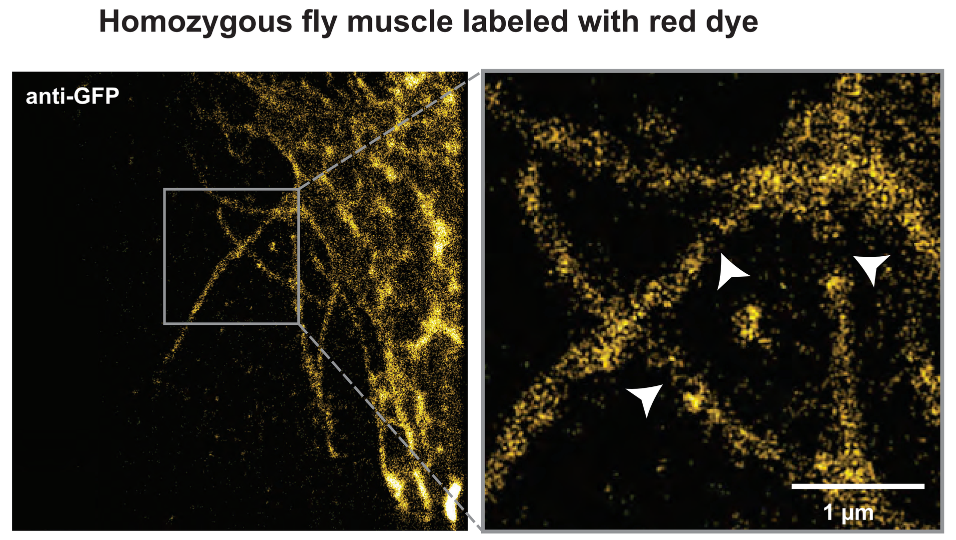



The heterozygous fly muscle 12 (;;*M12-Gal4, UAS-CD4::GFP/TM3*) aged at 15-17 hr AEL is stained with anti-GFP and farred dye-conjugated secondary antibodies. The arrowheads on the right panel show examples of discontinuous membrane labeling.

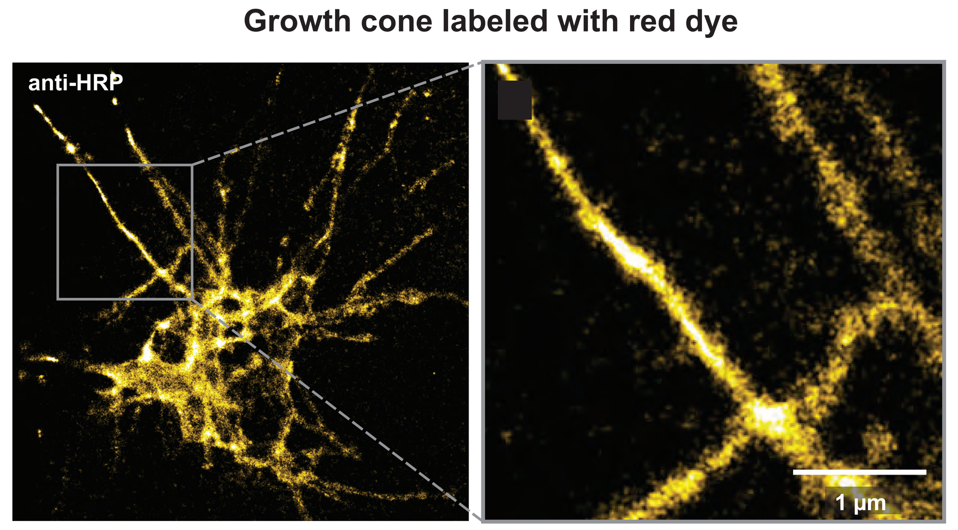



The homozygous fly (;;*M12-Gal4, UAS-CD4::GFP*) demonstrates improved labeling density of membrane on muscle 12 in embryos aged 15-17 hr AEL stained with the same conditions as heterozygous flies.The right panel demonstrates increased continuity on the membrane labeling.

## *A red dye as a second probe for dual color SMLM imaging*



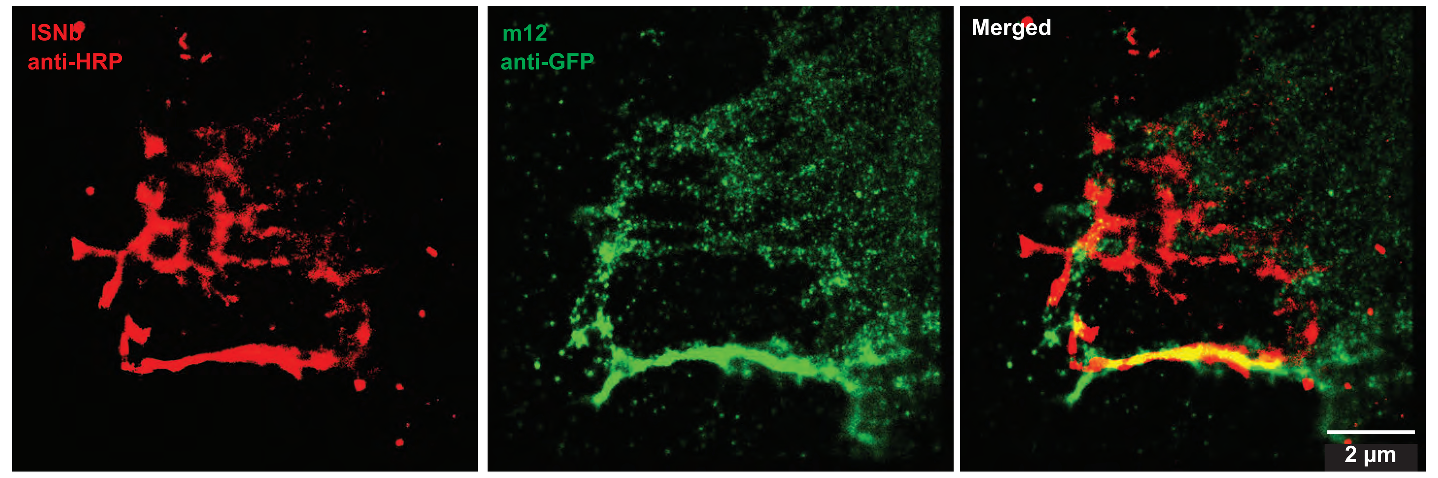

Homozgous fly muscle 12 (;;*M12-Gal4, UAS-CD4::GFP*) at age 15-17 hr AEL stained with red dye labeled secondary antibody against anti-GFP (left panel) has occasional gaps in membrane labeling shown by arrowheads (right panel).




The red-dye labeled anti-HRP works well for imaging the ISNb growth cone (15 hr AEL) with more continuous labeling (right panel).

## Initial test of 2-color imaging at the neuromuscular junction

### *Sequential two-channel SMLM image*




Two-color imaging of the neuromuscular junction at 15 AEL with the ISNB labeled with red dye and the muscle 12 labeled with the farred dye shows interaction of synaptic partners.

## Summary & Future Directions

With the goal of elucidating molecular recognition and recruitment mechanisms during synaptogenesis, we applied and optimized SMLM techniques to the fly embryo. Specifically, we used anti-HRP labeling on the presynaptic partner as a benchmark for optimizing labeling density and signal sensitivity which provides us insight into some morphological characteristics of filopodia. We also optimized muscle labeling for two-color SMLM imaging by increasing the genetic dosage and identifying a second dye. With two promising SMLM compatible dyes we took our first image at the NMJ.

In continuing our studies with these tools, we will investigate the changes in the filopodia structure to understand how these changes contribute to specificity of interaction between synaptic partners. We aim to furthermore investigate the distribution of synaptic proteins throughout the filopodia during partner interactions. Our investigations will provide insight into mechanisms of glutamatergic synapse formation at the early stages of the circuit development.

## Acknowledgements

We thank all members of the Kamiyama laboratory for the critical commentary on this project. This work was supported by NIH R01 NS107558 (to M.I., and D.K.), and an NSF 1350654 (to A.M.).